

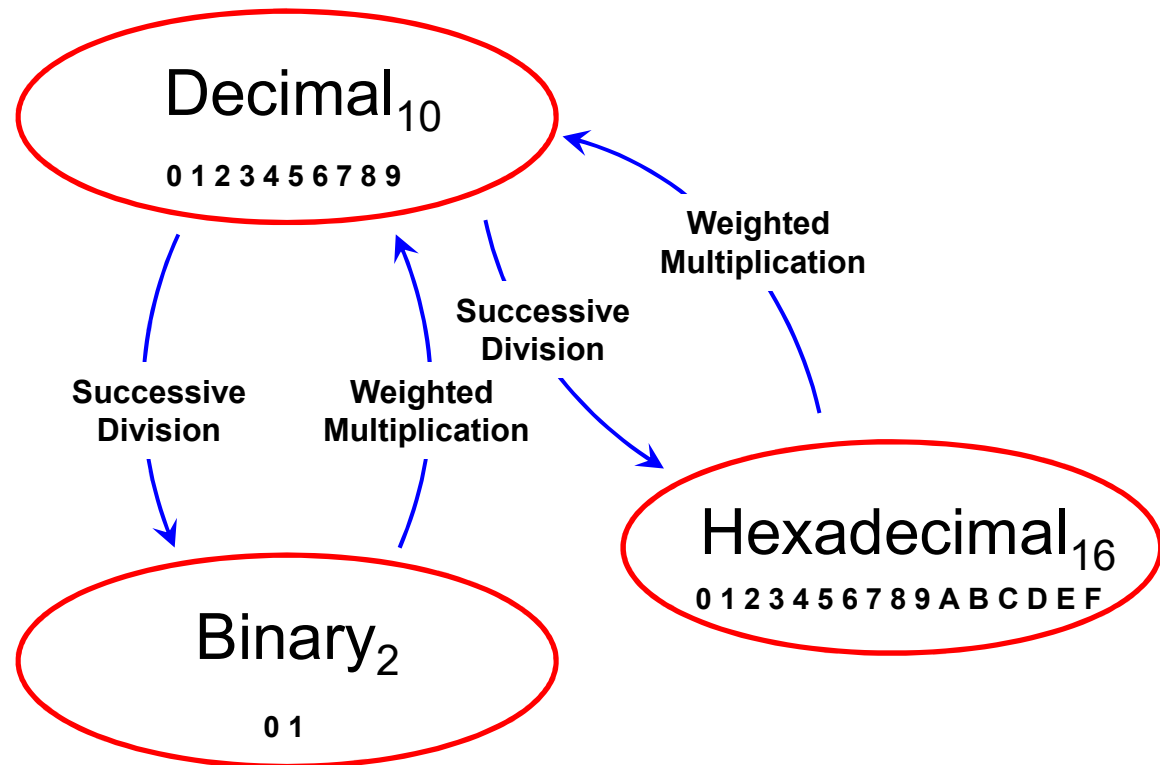
What, More Number Systems?

Why do we need more number systems?

- Humans understand decimal
- Computers understand binary only.
- Since computers have 32, 64, and even 128 bit busses, displaying numbers in binary is cumbersome.
- Data on a 32 bit data bus would look like the following:
0110 1001 0111 0001 0011 0100 1100 1010
- Hexadecimal (base 16) number systems is used to represent binary data in a more compact form.



Converting To and From Decimal



Counting . . . 2, 8, 10, 16

Decimal	Binary	Octal	Hexadecimal
0	00000	0	0
1	00001	1	1
2	00010	2	2
3	00011	3	3
4	00100	4	4
5	00101	5	5
6	00110	6	6
7	00111	7	7
8	01000	10	8
9	01001	11	9
10	01010	12	A
11	01011	13	B
12	01100	14	C
13	01101	15	D
14	01110	16	E
15	01111	17	F
16	10000	20	10
17	10001	21	11
18	10010	22	12
19	10011	23	13



Decimal $\leftrightarrow$ Hexadecimal Conversion

The Process: Successive Division

- Divide the decimal number by **16**; the remainder is the LSB of the **hexadecimal** number.
- If the quotient is zero, the conversion is complete. Otherwise repeat step (a) using the quotient as the decimal number. The new remainder is the next most significant bit of the **hexadecimal** number.

Example:

Convert the decimal number 94_{10} into its hexadecimal equivalent.

$$16 \overline{) 94} \quad \begin{array}{l} 5 \\ r = E \leftarrow \text{LSB} \end{array}$$

$$16 \overline{) 5} \quad \begin{array}{l} 0 \\ r = 5 \leftarrow \text{MSB} \end{array}$$

$$\therefore 94_{10} = 5E_{16}$$



Example: Dec $\rightarrow$ Hex

Example:

Convert the decimal number 429_{10} into its hexadecimal equivalent.



Example: Dec $\rightarrow$ Hex

Example:

Convert the decimal number 429_{10} into its hexadecimal equivalent.

Solution:

$$16 \overline{) 429} \quad \begin{array}{l} 26 \\ \hline \end{array} \quad r = D (13) \leftarrow \text{LSB}$$

$$16 \overline{) 26} \quad \begin{array}{l} 1 \\ \hline \end{array} \quad r = A (10)$$

$$16 \overline{) 1} \quad \begin{array}{l} 0 \\ \hline \end{array} \quad r = 1 \quad \leftarrow \text{MSB}$$

$$\therefore 429_{10} = 1AD_{16} = 1AD_H$$



Hexadecimal $\leftrightarrow$ Decimal Process

The Process: Weighted Multiplication

- Multiply each bit of the **hexadecimal** number by its corresponding bit-weighting factor (i.e., Bit-0 $\rightarrow 16^0=1$; Bit-1 $\rightarrow 16^1=16$; Bit-2 $\rightarrow 16^2=256$; etc.).
- Sum up all of the products in step (a) to get the decimal number.

Example:

Convert the octal number $5E_{16}$ into its decimal equivalent.

5	E	
16^1	16^0	
16	1	} Bit-Weighting Factors
80	14	
+ =		94_{10}

$$\therefore 5E_{16} = 94_{10}$$



Example: Hex $\rightarrow$ Dec

Example:

Convert the hexadecimal number $B2E_H$ into its decimal equivalent.



Example: Hex $\rightarrow$ Dec

Example:

Convert the hexadecimal number $B2E_H$ into its decimal equivalent.

Solution:

B	2	E	
16^2	16^1	16^0	
256	16	1	
2816	+	32	+
		14	=
2862_{10}			

$$\therefore B2E_H = 2862_{10}$$



Hex Decimal Addition

$$4A6_{16} + 1B3_{16} = 659_{16}$$

$$\begin{array}{r} 1 \quad \text{carry} \\ 4A6 = 1190_{10} \\ + 1B3 = 435_{10} \\ \hline 659 = 1625_{10} \end{array}$$

Hex Decimal Subtraction

$$4A6_{16} - 1B3_{16} = 2F3_{16}$$

$$\begin{array}{r} 16 \text{ borrow} \\ {}^3 4A6 = 1190_{10} \\ - 1B3 = 435_{10} \\ \hline 2F3 = 755_{10} \end{array}$$